King Fahd University of Petroleum & Minerals

Information & Computer Science Department

جامعة الملك فهد للبترول والمعادن
King Fahd University of Petroleum & Minerals

251-ICS-321: Database Systems

Project # 2

Campus Events Database

By:

Zahra Almadeh, ID:202254580

Fatimah Al Tawfiq, ID:202272340

Group: F22

13/12/2025

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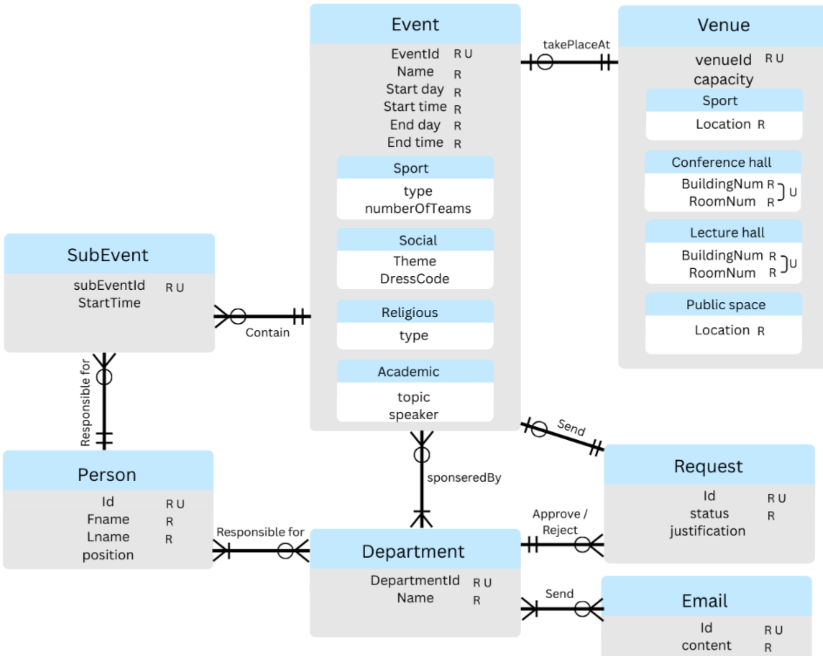
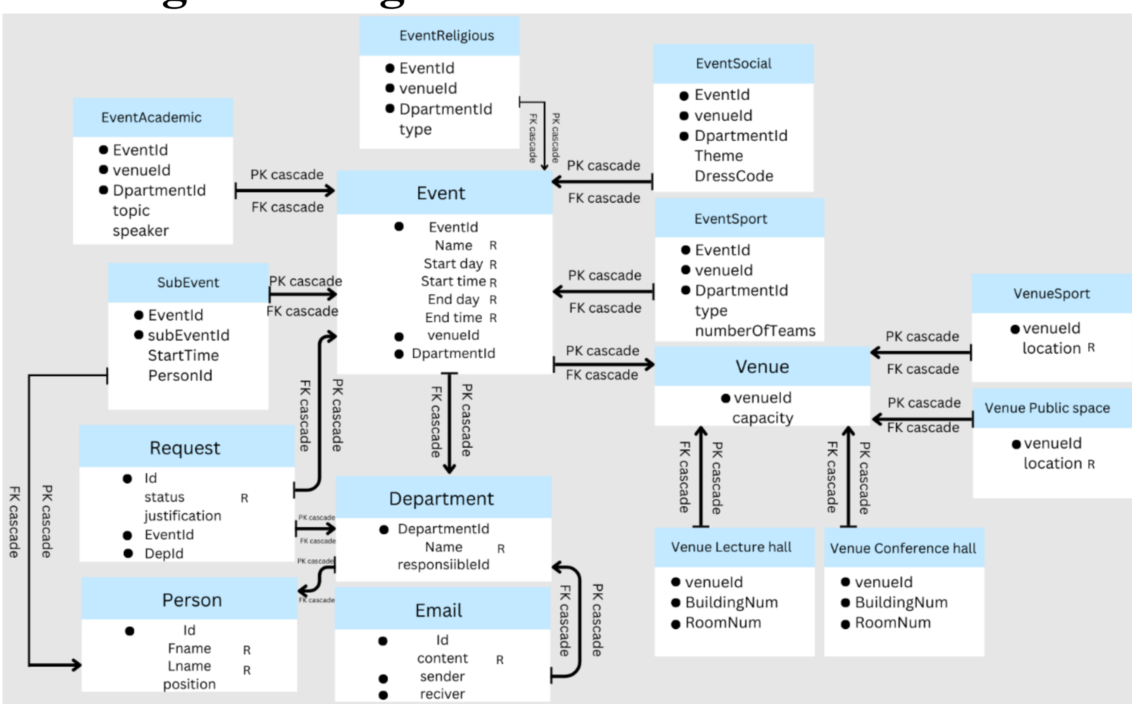
1. Conceptual Design:



2. Logical Design:



3. Normalized Schema:

1NF:

- **Event**
  Event(EventId, Name, StartDay, StartTime, EndDay, EndTime, venueId, DepartmentId)
- **EventAcademic**
  EventAcademic(EventId, venueId, DepartmentId, topic, speaker)
- **EventReligious**
  EventReligious(EventId, venueId, DepartmentId, type)
- **EventSocial**
  EventSocial(EventId, venueId, DepartmentId, Theme, DressCode)
- **EventSport**
  EventSport(EventId, venueId, DepartmentId, type, numberOfTeams)
- **SubEvent**
  SubEvent(EventId, subEventId, StartTime, PersonId)
- **Request**
  Request(Id, status, justification, EventId, DepId)
- **Department**
  Department(DepartmentId, Name, responsibleId)
- **Person**
  Person(Id, Fname, Lname, position)
- **Email**
  Email(Id, content, sender, receiver)
- **Venue**
  Venue(venueId,capacity)
- **VenueLectureHall**
  VenueLectureHall(venueId, BuildingNum, RoomNum)
- **VenueConferenceHall**
  VenueConferenceHall(venueId, BuildingNum, RoomNum)
- **VenueSport**
  VenueSport(venueId, location)
- **VenuePublicSpace**
  VenuePublicSpace(venueId, location)

2NF:

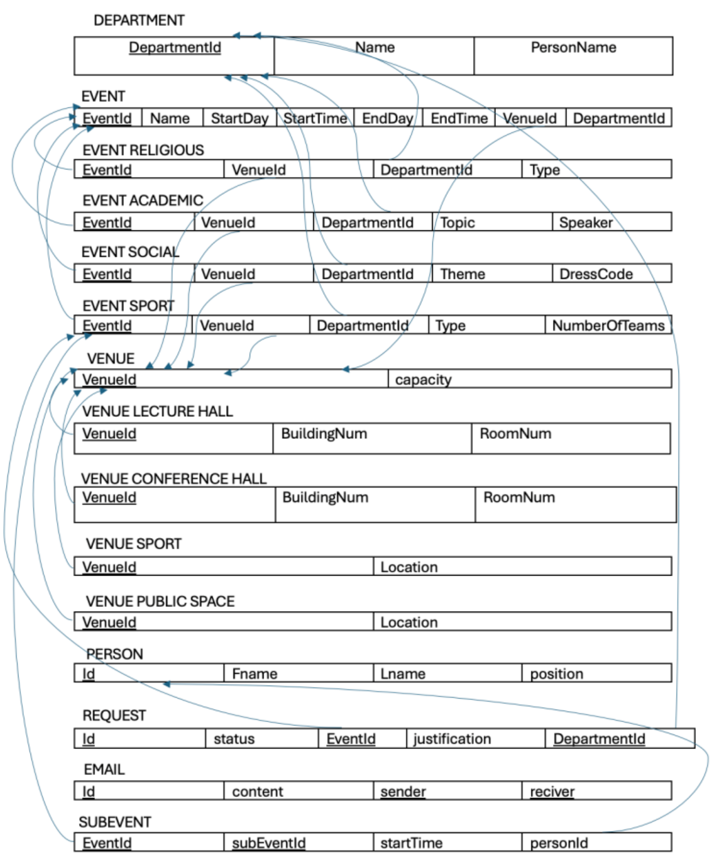
- **Event(EventId, Name, StartDay, StartTime, EndDay, EndTime, venueId, DepartmentId)**
  All non-key attrs depend on EventId.
- **EventAcademic(EventId, venueId, DepartmentId, topic, speaker)**
  All non-key attrs depend on EventId.
- **EventReligious(EventId, venueId, DepartmentId, type)**
  All non-key attrs depend on EventId.
- **EventSocial(EventId, venueId, DepartmentId, Theme, DressCode)**
  All non-key attrs depend on EventId.
- **EventSport(EventId, venueId, DepartmentId, type, numberOfTeams)**
  All non-key attrs depend on EventId.

- **SubEvent(EventId, subEventId, StartTime, PersonId)**
  (EventId, subEventId) is the key, and StartTime & PersonId depend on the whole pair, not on just one part.
- **Request(Id, status, justification, EventId, DepId)**
  All non-key attrs depend on Id.
- **Department(DepartmentId, Name, responsibleId)**
  All non-key attrs depend on DepartmentId.
- **Person(Id, Fname, Lname, position)**
  All non-key attrs depend on Id.
- **Email(Id, content, sender, receiver)**
  All non-key attrs depend on Id.
- **Venue(venueId,capacity)**
  All non-key attrs depend on venueId.
- **VenueLectureHall(venueId, BuildingNum, RoomNum)**
  Non-key attrs depend on venueId.
- **VenueConferenceHall(venueId, BuildingNum, RoomNum)**
  Non-key attrs depend on venueId.
- **VenueSport(venueId, location)**
  Non-key attrs depend on venueId.
- **VenuePublicSpace(venueId, location)**
  Non-key attrs depend on venueId.

BCNF:

- **Event**
  EventId → Name, StartDay, StartTime, EndDay, EndTime, venueId, DepartmentId
  key → **BCNF**.
- **EventAcademic**
  EventId → venueId, DepartmentId, topic, speaker
  key → **BCNF**.
- **EventReligious**
  EventId → venueId, DepartmentId, type
  key → **BCNF**.
- **EventSocial**
  EventId → venueId, DepartmentId, Theme, DressCode
  key → **BCNF**.
- **EventSport**
  EventId → venueId, DepartmentId, type, numberOfTeams
  key → **BCNF**.
- **SubEvent**
  (EventId, subEventId) → StartTime, PersonId
  (composite) key → **BCNF**.
- **Request**
  Id → status, justification, EventId, DepId
  key → **BCNF**.
- **Department**
  DepartmentId → Name, responsibleId
  key → **BCNF**.

- **Person**
  Id → Fname, Lname, position
  key → **BCNF**.
- **Email**
  Id → content, sender, receiver
  key → **BCNF**.
- **Venue**
  No non-trivial FD besides venueId → capacity →key determinant →**BCNF**.
- **VenueLectureHall**
  venueId → BuildingNum, RoomNum → key determinant → **BCNF**.
- **VenueConferenceHall**
  venueId → BuildingNum, RoomNum → key → **BCNF**.
- **VenueSport**
  venueId → location → key → **BCNF**.
- **VenuePublicSpace**
  venueId → location → key → **BCNF**.



4. Tools and Resources:

1. Canva: Creating Designs.
2. SQL Workbench: creating and testing the tables.

5. Tasks Division:

Task	Member
Conceptual Design	Fatimah
Logical Design	Fatimah
Normalized Schema	Zahra
SQL Queries	Zahra & Fatimah
Testing	Zahra

6. SQL Queries:

```
DELIMITER $$


CREATE PROCEDURE TablesCreation ()

BEGIN


CREATE TABLE VENUE(

   venueId varchar(3) NOT NULL PRIMARY KEY,

   capacity INTEGER

);


INSERT INTO VENUE VALUES

('s1',20), ('s2',37), ('s3',30), ('s4',25), ('s5',17),

('p1',20), ('p2',50), ('p3',30), ('p4',50), ('p5',20),

('c1',76), ('c2',27), ('c3',65), ('c4',75), ('c5',40),

('l1',20), ('l2',27), ('l3',30), ('l4',25), ('l5',17);


CREATE TABLE SportVenue(

   venueId varchar(3) NOT NULL PRIMARY KEY,

   location varchar(30) NOT NULL,

   FOREIGN KEY (venueId) REFERENCES VENUE(venueId) ON DELETE CASCADE

);


INSERT INTO SportVenue VALUES

('s1','B11'),
```

```sql
('s2','B36'),
('s3','B406'),
('s4','B407'),
('s5','B11');

CREATE TABLE PublicVenue(
    venueId varchar(3) NOT NULL PRIMARY KEY,
    location varchar(30) NOT NULL,
    FOREIGN KEY (venueId) REFERENCES VENUE(venueId) ON DELETE CASCADE
);

INSERT INTO PublicVenue VALUES
('p1','Student Park'),
('p2','Square'),
('p3','Student mall'),
('p4','Dhahran mosque'),
('p5','std restaurant');

CREATE TABLE ConfVenue(
    venueId varchar(3) NOT NULL,
    building varchar(5) NOT NULL,
    room varchar(5) NOT NULL,
    PRIMARY KEY (venueId, building, room),
    UNIQUE(building, room),
    FOREIGN KEY (venueId) REFERENCES VENUE(venueId) ON DELETE CASCADE
);

INSERT INTO ConfVenue VALUES
('c1','B6','125'),
('c2','B14','110'),
('c3','B59','1000'),
```

```sql
('c4','B14','108'),
('c5','B24','150');

CREATE TABLE LectVenue(
   venueId varchar(3) NOT NULL,
   building varchar(5) NOT NULL,
   room varchar(5) NOT NULL,
   PRIMARY KEY (venueId, building, room),
   UNIQUE(building, room),
   FOREIGN KEY (venueId) REFERENCES VENUE(venueId) ON DELETE CASCADE
);

INSERT INTO LectVenue VALUES
('l1','B7','100'),
('l2','B22','130'),
('l3','B59','1009'),
('l4','B59','1004'),
('l5','B24','150');

CREATE TABLE Person (
   person_id INT AUTO_INCREMENT PRIMARY KEY,
   fname VARCHAR(50) NOT NULL,
   lname VARCHAR(50) NOT NULL,
   position VARCHAR(20) NOT NULL,
   CHECK (position IN ('FACULTY','STUDENT','STAFF','DEPENDENT'))
);

INSERT INTO Person (person_id, fname, lname, position) VALUES
(1, 'Alice',  'Smith',  'FACULTY'),
(2, 'Bob',    'Jones',  'STUDENT'),
(3, 'Carol',  'Lee',    'STAFF'),
```

```
(4, 'David',  'Kim',    'FACULTY'),
(5, 'Emma',   'Brown',  'DEPENDENT');

CREATE TABLE DEPARTMENT(
   DepId varchar(5) NOT NULL PRIMARY KEY,
   DepName varchar(20) NOT NULL,
   responsibleId INT,
   FOREIGN KEY (responsibleId) REFERENCES Person(person_id) ON DELETE CASCADE
);

INSERT INTO DEPARTMENT VALUES
('d1','Management',1),
('d2','maintenance',2),
('d3','office services',3),
('d4','security',4),
('d5','IT',5);

CREATE TABLE Email(
   Id INT NOT NULL AUTO_INCREMENT PRIMARY KEY,
   content varchar(150) NOT NULL,
   sender varchar(5),
   reciver varchar(5),
   FOREIGN KEY (sender) REFERENCES DEPARTMENT(DepId) ON DELETE CASCADE,
   FOREIGN KEY (reciver) REFERENCES DEPARTMENT(DepId) ON DELETE CASCADE
);

INSERT INTO Email VALUES
(1,'Event approved','d1','d2'),
(2,'Event approved','d1','d3'),
(3,'Event approved','d1','d4'),
(4,'Event approved','d1','d5'),
```

```
(5,'Event approved','d1','d2');

CREATE TABLE Event (
   event_id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(200) NOT NULL,
   start_time DATETIME NOT NULL,
   end_time DATETIME NOT NULL,
   venue_id varchar(3) NOT NULL,
   department_id varchar(5) NOT NULL,
   event_type VARCHAR(20) NOT NULL,
   CONSTRAINT chk_event_time_order CHECK (end_time > start_time),
   CONSTRAINT chk_event_max_duration CHECK (TIMESTAMPDIFF(DAY, start_time, end_time) <= 3),
   CONSTRAINT chk_event_start_time CHECK (TIME(start_time) >= '08:00:00'),
   CONSTRAINT chk_event_end_time CHECK (TIME(end_time) <= '23:59:59'),
   FOREIGN KEY (venue_id) REFERENCES VENUE(venueId) ON DELETE CASCADE,
   FOREIGN KEY (department_id) REFERENCES DEPARTMENT(DepId) ON DELETE CASCADE
);

CREATE TABLE EventReligious (
   event_id INT PRIMARY KEY,
   rel_type VARCHAR(100),
   FOREIGN KEY (event_id) REFERENCES Event(event_id) ON DELETE CASCADE
);

CREATE TABLE EventAcademic (
   event_id INT PRIMARY KEY,
   topic VARCHAR(200),
   speaker VARCHAR(200),
   FOREIGN KEY (event_id) REFERENCES Event(event_id) ON DELETE CASCADE
);
```

```
CREATE TABLE EventSocial (
    event_id INT PRIMARY KEY,
    theme VARCHAR(200),
    dresscode VARCHAR(100),
    FOREIGN KEY (event_id) REFERENCES Event(event_id) ON DELETE CASCADE
);

CREATE TABLE EventSport (
    event_id INT PRIMARY KEY,
    sport_type VARCHAR(100),
    number_of_teams INT,
    FOREIGN KEY (event_id) REFERENCES Event(event_id) ON DELETE CASCADE
);

INSERT INTO Event (event_id, name, start_time, end_time, venue_id, department_id, event_type) VALUES
(1, 'Jumu'ah Prayer Program',      '2025-03-01 12:30:00', '2025-03-01 14:00:00', 's1', 'd1', 'RELIGIOUS'),
(2, 'Taraweeh Night Program',      '2025-03-01 20:00:00', '2025-03-01 22:00:00', 's2', 'd1', 'RELIGIOUS'),
(3, 'Islamic Lecture (Dars)',      '2025-03-02 09:30:00', '2025-03-02 11:00:00', 's1', 'd2', 'RELIGIOUS'),
(4, 'Qur'an Recitation Circle',    '2025-03-03 16:00:00', '2025-03-03 18:00:00', 's3', 'd2', 'RELIGIOUS'),
(5, 'Eid Celebration Program',     '2025-03-04 19:00:00', '2025-03-04 21:00:00', 's3', 'd3', 'RELIGIOUS'),
(6, 'AI in Education Seminar',     '2025-03-05 10:00:00', '2025-03-05 12:00:00', 's4', 'd1', 'ACADEMIC'),
(7, 'Chemistry Research Colloquium','2025-03-05 14:00:00', '2025-03-05 16:00:00', 's4', 'd2', 'ACADEMIC'),
(8, 'Mathematics Workshop',        '2025-03-06 09:00:00', '2025-03-06 11:30:00', 's4', 'd3', 'ACADEMIC'),
(9, 'Thesis Writing Clinic',       '2025-03-06 13:00:00', '2025-03-06 15:00:00', 's4', 'd3', 'ACADEMIC'),
(10,'Undergraduate Research Fair', '2025-03-07 10:00:00', '2025-03-07 14:00:00', 's5', 'd1', 'ACADEMIC'),
(11,'Spring Welcome Party',        '2025-03-08 18:00:00', '2025-03-08 22:00:00', 'p1', 'd4', 'SOCIAL'),
(12,'Board Games Night',           '2025-03-09 19:00:00', '2025-03-09 22:00:00', 'p1', 'd4', 'SOCIAL'),
(13,'Cultural Food Festival',      '2025-03-10 17:00:00', '2025-03-10 22:30:00', 'p2', 'd2', 'SOCIAL'),
(14,'Movie Under the Stars',       '2025-03-11 20:00:00', '2025-03-11 23:30:00', 'p2', 'd3', 'SOCIAL'),
(15,'Talent Show',                 '2025-03-12 18:00:00', '2025-03-12 21:30:00', 'p1', 'd1', 'SOCIAL'),
(16,'Interclass Football Match',  '2025-03-13 09:00:00', '2025-03-13 11:00:00', 's5', 'd5', 'SPORT'),
```

```sql
(17,'Basketball Tournament',      '2025-03-14 15:00:00', '2025-03-14 19:00:00', 's5', 'd5', 'SPORT'),
(18,'Tennis Doubles Day',         '2025-03-15 10:00:00', '2025-03-15 13:00:00', 's5', 'd5', 'SPORT'),
(19,'Campus Fun Run',             '2025-03-16 08:30:00', '2025-03-16 11:30:00', 's4', 'd3', 'SPORT'),
(20,'Volleyball Friendly',        '2025-03-17 16:00:00', '2025-03-17 18:30:00', 's4', 'd4', 'SPORT');

INSERT INTO EventReligious (event_id, rel_type) VALUES
(1, 'Jumu’ah Prayer'),
(2, 'Taraweeh Night'),
(3, 'Islamic Lecture (Dars)'),
(4, 'Qur’an Recitation Circle'),
(5, 'Eid Celebration Program');

INSERT INTO EventAcademic (event_id, topic, speaker) VALUES
(6,  'AI in Modern Classrooms',        'Dr. Alice Smith'),
(7,  'Green Chemistry Approaches',     'Dr. David Kim'),
(8,  'Problem Solving in Algebra',     'Prof. Carol Lee'),
(9,  'Academic Writing Techniques',    'Dr. Emma Brown'),
(10, 'Student Research Presentations', 'Prof. Bob Jones');

INSERT INTO EventSocial (event_id, theme, dresscode) VALUES
(11, 'Spring Welcome',      'Casual'),
(12, 'Game Night',          'Casual'),
(13, 'World Cultures',      'Traditional / Cultural'),
(14, 'Outdoor Cinema',      'Comfortable'),
(15, 'Show Your Talent',    'Smart Casual');

INSERT INTO EventSport (event_id, sport_type, number_of_teams) VALUES
(16, 'Football',   2),
(17, 'Basketball', 4),
(18, 'Tennis',     8),
(19, 'Running',    3),
```

```
(20, 'Volleyball', 2);

CREATE TABLE Request (
    request_id INT AUTO_INCREMENT PRIMARY KEY,
    event_id INT NOT NULL,
    department_id varchar(5) NOT NULL,
    status VARCHAR(20),
    justification TEXT,
    UNIQUE (event_id),
    FOREIGN KEY (event_id) REFERENCES Event(event_id) ON DELETE CASCADE,
    FOREIGN KEY (department_id) REFERENCES DEPARTMENT(DepId) ON DELETE CASCADE
);

INSERT INTO Request (request_id, event_id, department_id, status, justification) VALUES
(1,  1,  'd1', 'APPROVED', NULL),
(2,  2,  'd1', 'PENDING',  NULL),
(3,  3,  'd2', 'REJECTED', 'Conflicts with another major campus event'),
(4,  6,  'd1', 'APPROVED', NULL),
(5, 11,  'd4', 'APPROVED', NULL),
(6, 16,  'd5', 'PENDING',  NULL),
(7, 17,  'd5', 'APPROVED', NULL),
(8, 13,  'd2', 'REJECTED', 'Venue unavailable on requested date');

CREATE TABLE SubEvent (
    event_id INT NOT NULL,
    subevent_id INT NOT NULL,
    start_time DATETIME NOT NULL,
    allocated_minutes INT NOT NULL,
    person_id INT NOT NULL,
    PRIMARY KEY (event_id, subevent_id),
    FOREIGN KEY (event_id) REFERENCES Event(event_id) ON DELETE CASCADE,
```

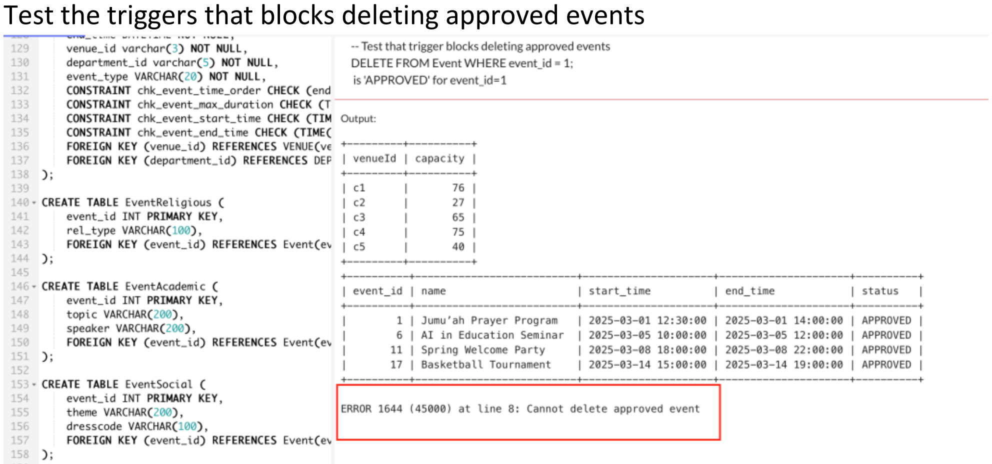
```sql
    FOREIGN KEY (person_id) REFERENCES Person(person_id)
);

INSERT INTO SubEvent (event_id, subevent_id, start_time, allocated_minutes, person_id) VALUES
(6,  1, '2025-03-05 10:00:00', 45, 1),
(6,  2, '2025-03-05 10:45:00', 45, 4),
(11, 1, '2025-03-08 18:00:00', 60, 2),
(11, 2, '2025-03-08 19:00:00', 90, 3),
(16, 1, '2025-03-13 09:00:00', 30, 5),
(16, 2, '2025-03-13 09:30:00', 60, 1);

END $$

DELIMITER ;
CALL TablesCreation();

-- Trigger to block deletion of approved events
DELIMITER $$
CREATE TRIGGER block_delete_event
BEFORE DELETE ON Event
FOR EACH ROW
BEGIN
  IF ((SELECT status FROM Request WHERE event_id = OLD.event_id) = 'APPROVED') THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Cannot delete approved event';
  END IF;
END $$
DELIMITER ;

-- Trigger to ensure subevent start time is not before Event start_time
DELIMITER $$
CREATE TRIGGER check_subevent_start
```

```
BEFORE INSERT ON SubEvent
FOR EACH ROW
BEGIN
  IF (NEW.start_time < (SELECT start_time FROM Event WHERE event_id = NEW.event_id)) THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'SubEvent too early';
  END IF;
END $$
DELIMITER ;

-- Trigger to ensure subevent start time is not after Event end_time
DELIMITER $$
CREATE TRIGGER check_subevent_end
BEFORE INSERT ON SubEvent
FOR EACH ROW
BEGIN
  IF (NEW.start_time > (SELECT end_time FROM Event WHERE event_id = NEW.event_id)) THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'SubEvent too late';
  END IF;
END $$
DELIMITER ;

-- Trigger to insert email notifications on approval
DELIMITER $$
CREATE TRIGGER notify_on_approval
AFTER UPDATE ON Request
FOR EACH ROW
BEGIN
  IF (NEW.status = 'APPROVED') THEN
    INSERT INTO Email (content, sender, reciver) VALUES
    ('Event approved', 'd1', 'd2'),
    ('Event approved', 'd1', 'd3'),
```

```
    ('Event approved', 'd1', 'd4'),
    ('Event approved', 'd1', 'd5');
  END IF;
END $$
DELIMITER ;

CREATE VIEW ApprovedEvents AS
SELECT e.event_id, e.name, e.start_time, e.end_time, r.status
FROM Event e
JOIN Request r ON r.event_id = e.event_id
WHERE r.status = 'APPROVED';
CREATE OR REPLACE VIEW EMAILSENDT AS
SELECT E.content, S.DepName AS sender_name, R.DepName AS receiver_name
FROM Email E
JOIN DEPARTMENT S ON E.sender = S.DepId
JOIN DEPARTMENT R ON E.reciver = R.DepId;
UPDATE Request
SET status = 'APPROVED'
WHERE request_id = 2;
```

7. Test

1- Test the triggers that blocks deleting approved events



2- Test the constraint “chk_event_start_time”

```sql
    venue_id varchar(3) NOT NULL,
    department_id varchar(5) NOT NULL,
    event_type VARCHAR(20) NOT NULL,
    CONSTRAINT chk_event_time_order CHECK (end
    CONSTRAINT chk_event_max_duration CHECK (T
    CONSTRAINT chk_event_start_time CHECK (TIM
    CONSTRAINT chk_event_end_time CHECK (TIME(
    FOREIGN KEY (venue_id) REFERENCES VENUE(ve
    FOREIGN KEY (department_id) REFERENCES DEP
);

CREATE TABLE EventReligious (
    event_id INT PRIMARY KEY,
    rel_type VARCHAR(100),
    FOREIGN KEY (event_id) REFERENCES Event(ev
);

CREATE TABLE EventAcademic (
    event_id INT PRIMARY KEY,
    topic VARCHAR(200),
    speaker VARCHAR(200),
    FOREIGN KEY (event_id) REFERENCES Event(ev
);

CREATE TABLE EventSocial (
    event_id INT PRIMARY KEY,
    theme VARCHAR(200),
    dresscode VARCHAR(100),
    FOREIGN KEY (event_id) REFERENCES Event(ev
);
```

```sql
INSERT INTO SubEvent (event_id, subevent_id, start_time, allocated_minutes, person_id)
VALUES (1, 99, '2025-03-01 07:00:00', 30, 1);
```

Output:

```
+---------+----------+
| venueId | capacity |
+---------+----------+
| c1      |       76 |
| c2      |       27 |
| c3      |       65 |
| c4      |       75 |
| c5      |       40 |
+---------+----------+
+----------+-----------------------+---------------------+---------------------+----------+
| event_id | name                  | start_time          | end_time            | status   |
+----------+-----------------------+---------------------+---------------------+----------+
|        1 | Jumu'ah Prayer Program | 2025-03-01 12:30:00 | 2025-03-01 14:00:00 | APPROVED |
|        6 | AI in Education Seminar | 2025-03-05 10:00:00 | 2025-03-05 12:00:00 | APPROVED |
|       11 | Spring Welcome Party  | 2025-03-08 18:00:00 | 2025-03-08 22:00:00 | APPROVED |
|       17 | Basketball Tournament | 2025-03-14 15:00:00 | 2025-03-14 19:00:00 | APPROVED |
+----------+-----------------------+---------------------+---------------------+----------+

ERROR 1644 (45000) at line 7: SubEvent too early
```

3- Test the constraint “chk_event_max_duration”

```sql
    venue_id varchar(3) NOT NULL,
    department_id varchar(5) NOT NULL,
    event_type VARCHAR(20) NOT NULL,
    CONSTRAINT chk_event_time_order CHECK (end
    CONSTRAINT chk_event_max_duration CHECK (T
    CONSTRAINT chk_event_start_time CHECK (TIM
    CONSTRAINT chk_event_end_time CHECK (TIME(
    FOREIGN KEY (venue_id) REFERENCES VENUE(ve
    FOREIGN KEY (department_id) REFERENCES DEP
);

CREATE TABLE EventReligious (
    event_id INT PRIMARY KEY,
    rel_type VARCHAR(100),
    FOREIGN KEY (event_id) REFERENCES Event(ev
);

CREATE TABLE EventAcademic (
    event_id INT PRIMARY KEY,
    topic VARCHAR(200),
    speaker VARCHAR(200),
    FOREIGN KEY (event_id) REFERENCES Event(ev
);

CREATE TABLE EventSocial (
    event_id INT PRIMARY KEY,
    theme VARCHAR(200),
    dresscode VARCHAR(100),
    FOREIGN KEY (event_id) REFERENCES Event(ev
);

CREATE TABLE EventSport (
    event_id INT PRIMARY KEY,
    sport_type VARCHAR(100),
    number_of_teams INT,
    FOREIGN KEY (event_id) REFERENCES Event(ev
);

INSERT INTO Event (event_id, name, start_time,
```

```sql
INSERT INTO Event(name,start_time,end_time,venue_id,department_id,event_type)
VALUES ('Long event','2025-03-01 08:00:00','2025-03-05 08:00:00','s1','d1','SOCIAL');
```

Output:

```
+---------+----------+
| venueId | capacity |
+---------+----------+
| c1      |       76 |
| c2      |       27 |
| c3      |       65 |
| c4      |       75 |
| c5      |       40 |
+---------+----------+
+----------+-----------------------+---------------------+---------------------+----------+
| event_id | name                  | start_time          | end_time            | status   |
+----------+-----------------------+---------------------+---------------------+----------+
|        1 | Jumu'ah Prayer Program | 2025-03-01 12:30:00 | 2025-03-01 14:00:00 | APPROVED |
|        6 | AI in Education Seminar | 2025-03-05 10:00:00 | 2025-03-05 12:00:00 | APPROVED |
|       11 | Spring Welcome Party  | 2025-03-08 18:00:00 | 2025-03-08 22:00:00 | APPROVED |
|       17 | Basketball Tournament | 2025-03-14 15:00:00 | 2025-03-14 19:00:00 | APPROVED |
+----------+-----------------------+---------------------+---------------------+----------+

ERROR 3819 (HY000) at line 7: Check constraint 'chk_event_max_duration' is violated.
```

4- Test that will successfully insert a valid SubEvent within the event time range

```sql
INSERT INTO SubEvent (event_id, subevent_id, start_time, allocated_minutes, person_id)
VALUES (6, 101, '2025-03-05 11:00:00', 30, 4);
```

Output:

```
+----------+-------------+---------------------+-------------------+-----------+
| event_id | subevent_id | start_time          | allocated_minutes | person_id |
+----------+-------------+---------------------+-------------------+-----------+
|        6 |           1 | 2025-03-05 10:00:00 |                45 |         1 |
|        6 |           2 | 2025-03-05 10:45:00 |                45 |         4 |
+----------+-------------+---------------------+-------------------+-----------+
```

before insert

```
+----------+-------------+---------------------+-------------------+-----------+
| event_id | subevent_id | start_time          | allocated_minutes | person_id |
+----------+-------------+---------------------+-------------------+-----------+
|        6 |           1 | 2025-03-05 10:00:00 |                45 |         1 |
|        6 |           2 | 2025-03-05 10:45:00 |                45 |         4 |
|        6 |         101 | 2025-03-05 11:00:00 |                30 |         4 |
+----------+-------------+---------------------+-------------------+-----------+
```

after insert

5- Test the email notifications after updating a request status to 'APPROVED'

Output:

```
+----------------+-------------+-----------------+
| content        | sender_name | receiver_name   |
+----------------+-------------+-----------------+
| Event approved | Management  | maintenance     |
| Event approved | Management  | office services |
| Event approved | Management  | security        |
| Event approved | Management  | IT              |
| Event approved | Management  | maintenance     |
+----------------+-------------+-----------------+
```

before approval

```
+----------------+-------------+-----------------+
| content        | sender_name | receiver_name   |
+----------------+-------------+-----------------+
| Event approved | Management  | maintenance     |
| Event approved | Management  | maintenance     |
| Event approved | Management  | maintenance     |
| Event approved | Management  | office services |
| Event approved | Management  | office services |
| Event approved | Management  | security        |
| Event approved | Management  | security        |
| Event approved | Management  | IT              |
| Event approved | Management  | IT              |
+----------------+-------------+-----------------+
```

after approval